



1. Introduction

This document is a hardware user guide for the VCP-G based on the TCC7045 application processor. This document describes system installation, debugging, and detailed information on overall design and usage of the VCP-G.

Table 1.1 describes the features of the VCP-G.

Table 1.1 Features of VCP-G

Part Name		TCC7045
Package		Package Pin to Pin Compatible FBGA 196-pin (12BD)
CPU Frequency		200 MHz (Up to 300 MHz)
On-chip Memory	Program Flash	4 MB
	SRAM	512 KB (Including Retention RAM 16 KB)
	DataFlash	256 KB
	DMA Channel	16 channels
Peripheral	Ethernet	1 Gbps with AVB
	CAN / CANFD	3-channel
	Dedicated LIN / UART	3 channels (Maximum 6 channels)
	Dedicated I2C	3 channels (Maximum 6 channels)
	Dedicated GPSB (SPI)	2 channels (Maximum 5 channels)

	MFIO (Allocated UART, I2C, GPSB)		3 channels
	ADC	Resolution	12-bit SAR type
		Channels	12-channel x 2 groups
		Input Range	3.3V
		Sample Rate	Over 1.0 MSPs
	I2S		1-channel
	Serial Flash Interface		Quad SPI
Power System			3.3V single
Temperature			-40 °C to 105 °C

1.1 Terminology

Table 1.2 Terminology

Terminology	Definition
ADC	Analog to Digital Converter
FWDN	Firmware Download
GPIO	General Purpose Input Output
MCU	Micro-controller Unit
TOPST	Total Open-Platform for System development and Training
VCP	Vehicle Control Processer

2. Block Diagram

2.1 System Block Diagram

Figure 2.1 shows the system block diagram of VCP-G.

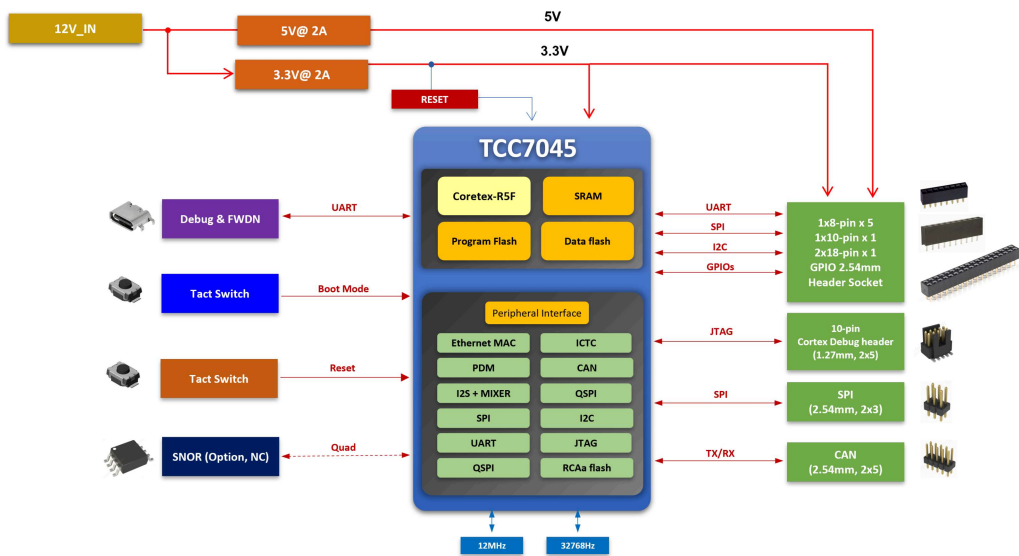


Figure 2.1 System Block Diagram

3. VCP-G Overview

The VCP-G can be used for the following purposes:

- System development
- Training

Table 3.1 describes the default configuration of the VCP-G.

Table 3.1 Default Configuration of VCP-G

Board Name	Description
TOPST_VCP_V2.1.1	MCU board for TOPST

3.1 VCP-G

Figure 3.1 shows the top view of VCP-G.

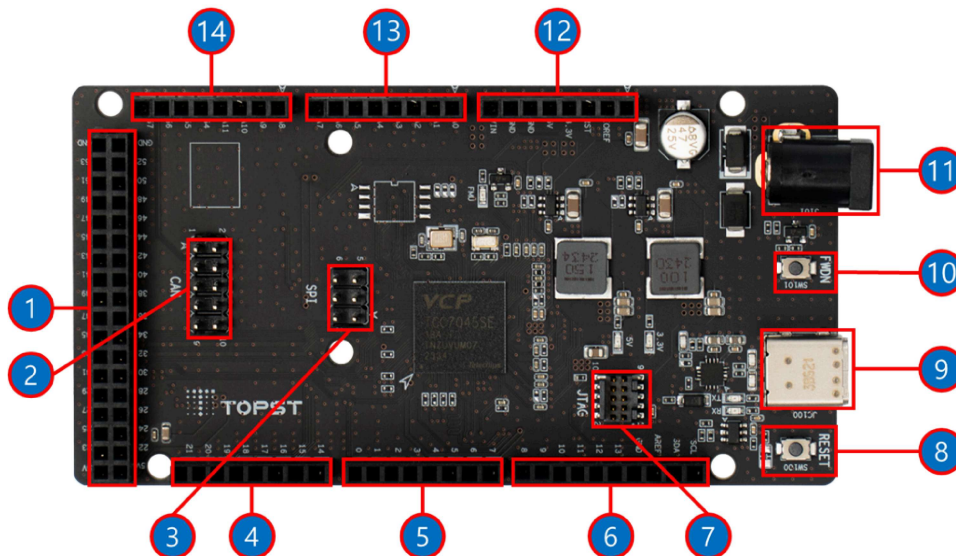


Figure 3.1 VCP-G (Top View)

Table 3.2 describes the connectors of VCP-G (top view).

Table 3.2 Connectors of VCP-G (Top View)

Number	Reference Number	Name	Description
1	J18D100	36-pin Female Header	Header for GPIO and ADC
2	J5D100	10-pin Male Header	Header for CAN
3	J3D100	6-pin Male Header	Header for SPI
4	J8D104	8-pin Female Header	Header for GPIO and ADC

5	J8D102	8-pin Female Header	Header for GPIO
6	J10D100	10-pin Female Header	Header for GPIO and ADC
7	J100	10-pin Male Header	Header for JTAG
8	SW100	RESET Tact Switch	GRESETn: Initializes the system and the power management of VCP-G
9	JC100	USB Type-C Connector	UART for debugging or FWDN port
10	SW101	Tact Switch	FWDN: Enter the Firmware download mode of VCP-G
11	J101	DC Jack	DC Power Input Jack
12	J8D100	8-pin Female Header	Header for Power and Reset
13	J8D101	8-pin Female Header	Header for GPIO and ADC
14	J8D103	8-pin Female Header	Header for GPIO and ADC

Figure 3.2 shows the bottom view of the VCP-G.

Note: Figure 3.2 currently shows the TOPST_VCP-G_V1.1.1 board. This image will be updated to the TOPST_VCP-G_V2.1.1 board.

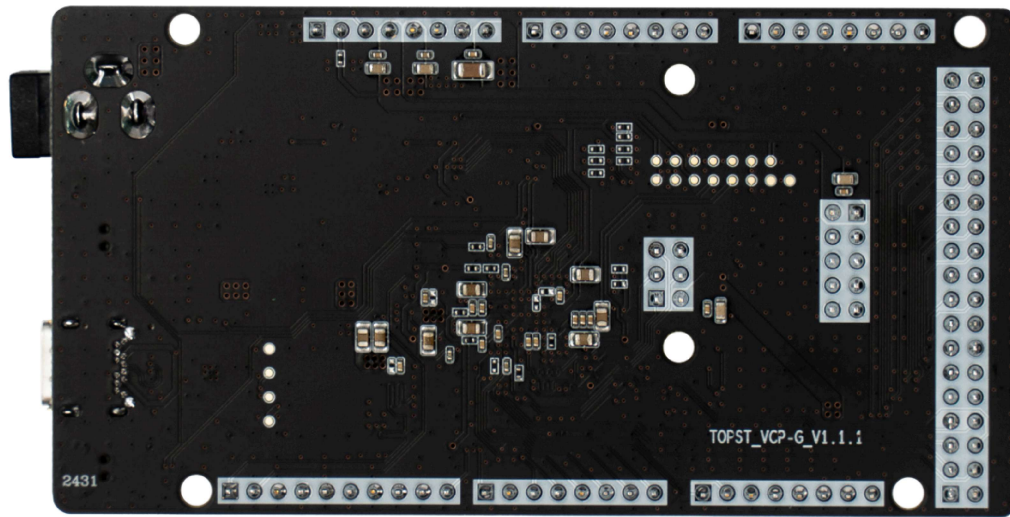


Figure 3.2 VCP-G (Bottom View)

4. Specifications

4.1 Quad SPI Flash Memory (U101)

The information on the Quad SPI flash memory is as follows:

- Density : 64 Mb

Note: SNOR is not mounted on the VCP-G by default.

4.2 Power-in Connector (J101)

DC 12V is supplied to the VCP-G through the DC jack of J101 from a 12V adaptor.

Figure 4.1 shows the location of J101.

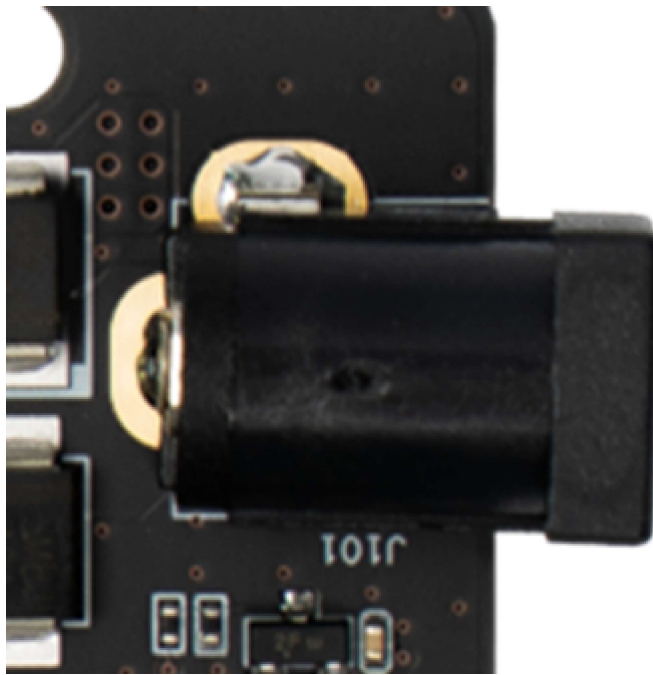


Figure 4.1 Power-in Connector (J101)

4.3 Connector for JTAG (J100)

A JTAG emulator can be connected to the VCP-G through J100 for debugging. Figure 4.2 shows the location of J100.

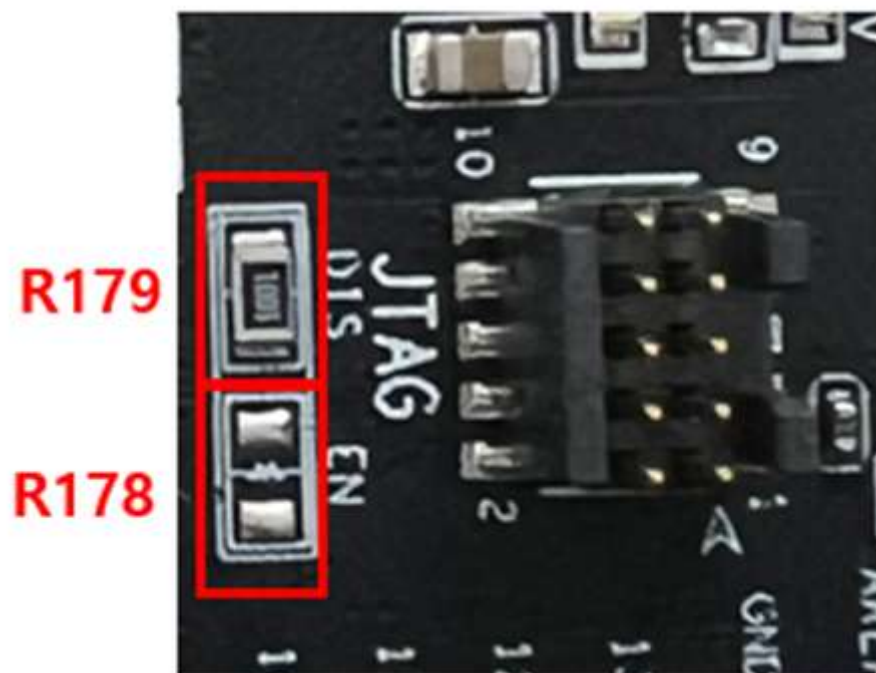


Figure 4.2 Connector for JTAG (J100)

JTAG is disabled by default. To enable JTAG, you must change the connections of R178 and R179. If TRSRn is set to high by R178, the MCU enters JTAG mode.

Table 4.1 describes the pins of J100.

Table 4.1 J100 Pin Description

Pin Number	Schematic Net Name	DIR	Description
		MCU ◀▶ J100	
1	SW_VDD_3P3	-	Power 3.3V
2	TMS	◀	Test Mode State
3	DGND	-	Ground
4	TCK	◀	Test Clock
5	DGND	-	Ground
6	TDO	▶	Test Data Output
7	NC	-	Not Connected
8	TDI	◀	Test Data In
9	DGND	-	Ground
10	JTAG_RESETh	◀	System Reset

Table 4.2 describes the setting of JTAG Disable/Enable.

Table 4.2 Setting of JTAG Disable/Enable

Mode	TRSTn Value	R178	R179
JTAG Disable (Default)	Low (1)	N.C	1K
JTAG Enable (Option)	High (1)	1K	N.C

4.4 FWDN Switch (SW101)

The VCP-G has one pin for boot configuration using Boot Mode (BM) and supports 2 modes: UART FWDN mode and normal mode.

Figure 4.3 shows the location of FWDN tact switch (SW101), which is used to select the boot mode of the VCP-G.

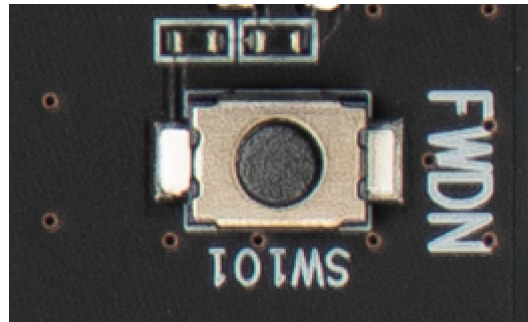


Figure 4.3 FWDN Tact Switch (SW101)

Table 4.3 describes how to use the FWDN tact switch (SW101) to select the boot mode.

Table 4.3 Description of Tact Switch (SW101) for Boot Mode

Mode	BM00 Value	SW101 Status
Normal (Default)	Low (1)	Default
FWDN (Option)	High (1)	Pushed and Power

4.4.1 FWDN Mode Method

There are two methods to enter FWDN mode as follows.

4.4.1.1 Method 1

While pressing the FWDN switch (SW101), connect the 12V power supply to turn on the VCP-G board.

The FWDN red indicator turns on when power is applied while the FWDN switch is pressed. After releasing the FWDN switch (SW101), the MCU enters FWDN mode.

Figure 4.4 shows how to enter FWDN mode by using method 1.

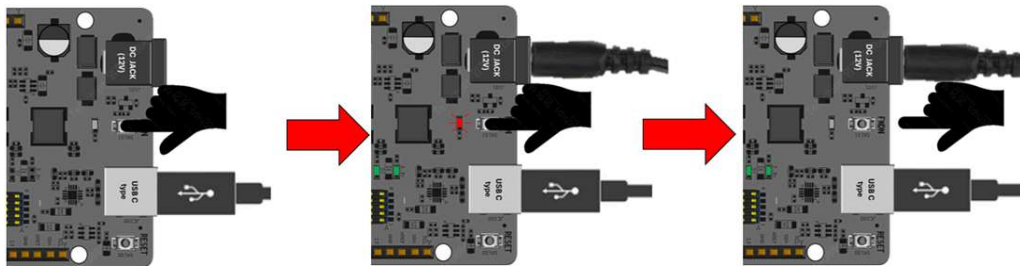


Figure 4.4 Enter FWDN Mode by using Method 1

4.4.1.2 Method 2

While the VCP-G board is connected to the 12V power supply, press the FWDN switch (SW101) and then press the RESET tact switch (SW100). The FWDN red indicator turns on when the power is applied while the FWDN switch is pressed. The 3.3V green indicator turns off while the RESET tact switch is pressed. After releasing the FWDN switch (SW101), the MCU enters FWDN mode.

Figure 4.5 shows the FWDN mode by using method 2.

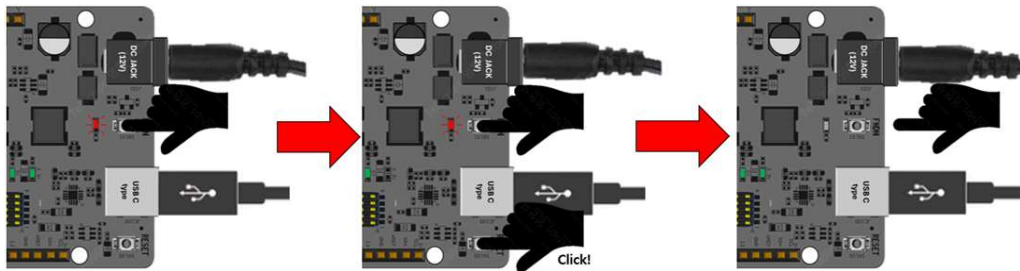


Figure 4.5 Enter FWDN Mode by using Method 2

4.5 RESET Tact Switch (SW100)

The VCP-G has one RESET switch for power of RESET using GRESETn pin. Figure 4.6 shows the RESET tact switch (SW100).

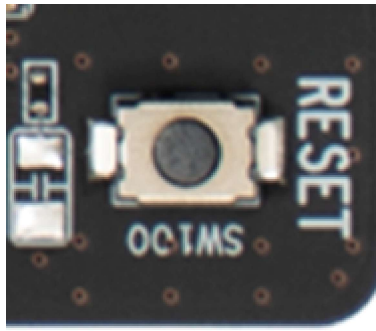


Figure 4.6 RESET Tact Switch (SW100)

4.5.1 RESET Tact Switch (SW100) Function

SW100 is a tact switch to reset the power block and system block in VCP-G.

The function of this button is as follows:

- Pressing the RESET tact switch (SW100) while the power is on forces the power block and the system of the VCP-G to reset.

Important: Be careful when pressing the tact switch as the power suddenly turns off and data may be corrupted.

4.6 Connector for Debugging and FWDN (JC100)

The JC100 is a standard USB Type-C connector. On the VCP-G, JC100 is used for debugging or FWDN through UART.

Figure 4.7 shows the location of JC100.

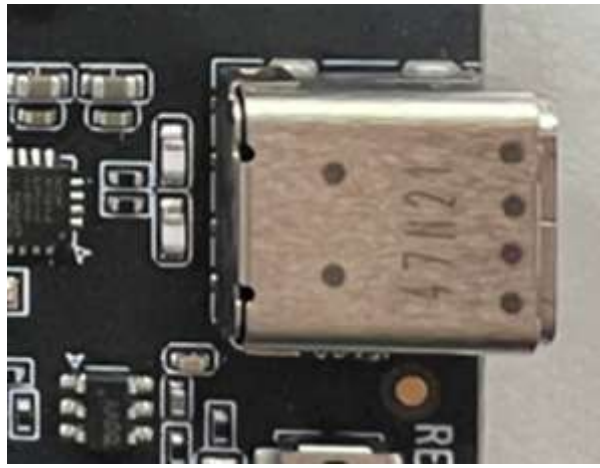


Figure 4.7 USB Type-C Connector (JC100)

You can perform FWDN or check debugging message of the VCP-G through JC100. JC100 on the VCP-G includes a built-in USB-to-UART bridge controller, so you can directly connect JC100 to PC by using the USB Type-C cable.

4.7 Pin Headers for GPIO, ADC, Power, CAN, and SPI

The VCP-G has nine 2.54 mm pin headers for power, GPIO, ADC, CAN, and SPI to connect to other peripherals such as sensors or sub-boards.

Table 4.4 describes the purpose of the nine pin headers on the VCP-G.

Table 4.4 Pin Headers on VCP-G

Number	Reference Number	Name	Description
1	J18D100	36-pin Female Header	Header for GPIO and ADC
2	J5D100	10-pin Male Header	Header for CAN
3	J3D100	6-pin Male Header	Header for SPI

4	J8D104	8-pin Female Header	Header for GPIO and ADC
5	J8D102	8-pin Female Header	Header for GPIO
6	J10D100	10-pin Female Header	Header for GPIO and ADC
7	J8D100	8-pin Female Header	Header for Power and Reset
8	J8D101	8-pin Female Header	Header for GPIO and ADC
9	J8D103	8-pin Female Header	Header for GPIO and ADC

Figure 4.8 shows the location of pin headers on the VCP-G.

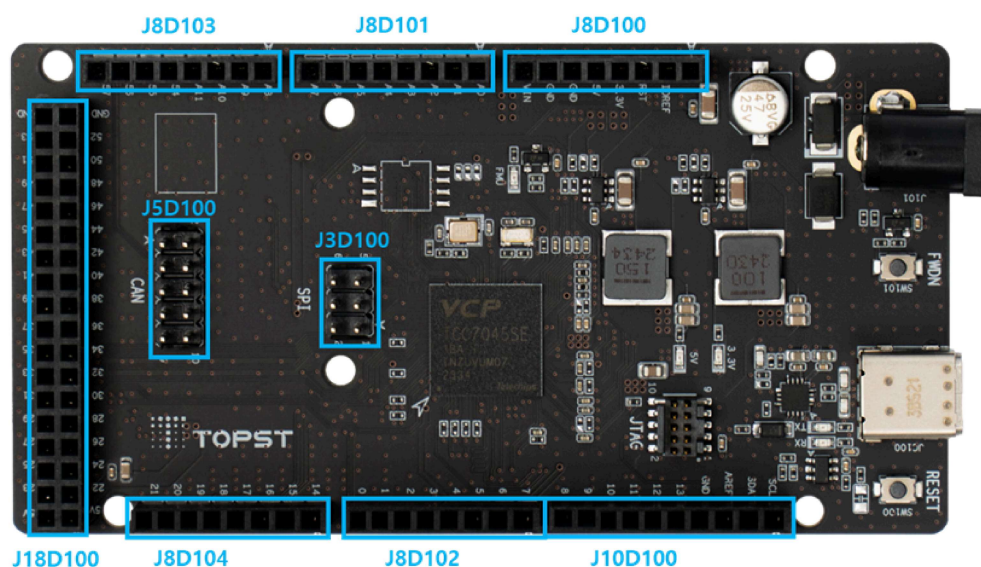


Figure 4.8 Pin Headers on VCP-G

Table 4.5 shows the pin description of J10D100.

Table 4.5 J10D100 Pin Description

Pin Number	J10D100			
	Port Name	Signal Name	DIR	Description
			MCU ◀▶ J10D100	

1	SCL	GPIO_AC07	◄►	GPIO or ADC signal
2	SDA	GPIO_AC06	◄►	GPIO or ADC signal
3	AREF	ADC06	◄	ADC signal
4	GND	DGND	-	Ground
5	13	GPIO_C12	◄►	GPIO signal
6	12	GPIO_C15	◄►	GPIO signal
7	11	GPIO_C14	◄►	GPIO signal
8	10	GPIO_C13	◄►	GPIO signal
9	9	GPIO_A12	◄►	GPIO signal
10	8	GPIO_B00	◄►	GPIO signal

Table 4.6 shows the pin description of J8D100.

Table 4.6 J8D100 Pin Description

Pin Number	J8D100			
	Port Name	Signal Name	DIR	Description
			MCU ◄► J8D100	
1	-	-	-	-
2	IOREF	VCP_3P3	-	Power 3.3V
3	RST	RESET	◄	Reset signal
4	3.3V	VCP_3P3	-	Power 3.3V
5	5V	VCP_5P0	-	Power 5.0V
6	GND	DGND	-	Ground
7	GND	DGND	-	Ground
8	VIN	VIN	-	Voltage input for VCP-G

Table 4.7 shows the pin description of J8D101.

Table 4.7 J8D101 Pin Description

Pin Number	J8D101			
	Port Name	Signal Name	DIR	Description
			MCU ◀▶ J8D101	
1	A0	ADC03	◀	ADC signal
2	A1	ADC04	◀	ADC signal
3	A2	GPIO_AC02	◀▶	GPIO signal
4	A3	GPIO_AC03	◀▶	GPIO signal
5	A4	GPIO_AC05	◀▶	GPIO signal
6	A5	GPIO_AC04	◀▶	GPIO signal
7	A6	ADC05	◀	ADC signal
8	A7	ADC01	◀	ADC signal

Table 4.8 shows the pin description of J8D102.

Table 4.8 J8D102 Pin Description

Pin Number	J8D102			
	Port Name	Signal Name	DIR	Description
			MCU ◀▶ J8D102	
1	7	GPIO_B01	◀▶	GPIO signal
2	6	GPIO_A13	◀▶	GPIO signal
3	5	GPIO_B10	◀▶	GPIO signal
4	4	GPIO_B27	◀▶	GPIO signal
5	3	GPIO_B11	◀▶	GPIO signal
6	2	GPIO_B28	◀▶	GPIO signal
7	1	GPIO_B25	◀▶	GPIO signal

8	0	GPIO_B26	◀▶	GPIO signal
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Table 4.9 shows the pin description of J8D103.

Table 4.9 J8D103 Pin Description

Pin Number	J8D103			
	Port Name	Signal Name	DIR	Description
			MCU ◀▶ J8D103	
1	A8	GPIO_AC08	◀▶	GPIO or ADC signal
2	A9	GPIO_AC09	◀▶	GPIO or ADC signal
3	A10	GPIO_AC10	◀▶	GPIO or ADC signal
4	A11	GPIO_ADC-2	◀	ADC signal
5	54	GPIO_K14	◀▶	GPIO signal
6	55	GPIO_K15	◀▶	GPIO signal
7	56	GPIO_K01	◀	GPIO signal
8	57	GPIO_K08	◀▶	GPIO signal

Table 4.10 shows the pin description of J8D104.

Table 4.10 J8D104 Pin Description

Pin Number	J8D104			
	Port Name	Signal Name	DIR	Description
			MCU ◀▶ J8D104	
1	14	GPIO_AC00	◀▶	GPIO or ADC signal

2	15	GPIO_AC01	◄►	GPIO or ADC signal
3	16	GPIO_A06	◄►	GPIO signal
4	17	GPIO_A07	◄►	GPIO signal
5	18	GPIO_A28	◄►	GPIO signal
6	19	GPIO_A29	◄►	GPIO signal
7	20	GPIO_B03	◄►	GPIO signal
8	21	GPIO_B02	◄►	GPIO signal

Table 4.11 shows the pin description of J3D100.

Table 4.11 J3D100 Pin Description

Pin Number	J3D100			
	Port Name	Signal Name	DIR	Description
			MCU ◄► J3D100	
1	MISO	GPIO_B07	◄►	GPIO signal
2	5V	VCP_5P0	-	Power 5.0V
3	SCK	GPIO_B04	◄►	GPIO signal
4	MOSI	GPIO_B06	◄►	GPIO signal
5	CMD	GPIO_B05	◄►	GPIO signal
6	GND	DGND	-	Ground

Table 4.12 shows the pin description of J18D100.

Table 4.12 J18D100 Pin Description

Pin Number	J18D100			
	Port Name	Signal Name	DIR	Description
			MCU ◄► J18D100	
1	5V	VCP_5P0	-	Power 5.0V

2	5V	VCP_5P0	-	Power 5.0V
3	22	GPIO_B24	◀▶	GPIO signal
4	23	GPIO_B23	◀▶	GPIO signal



8	27	GPIO_B19	◀▶	GPIO signal
9	28	GPIO_A30	◀▶	GPIO signal
10	29	GPIO_A27	◀▶	GPIO signal
11	230	GPIO_A26	◀▶	GPIO signal
12	31	GPIO_A24	◀▶	GPIO signal
13	32	GPIO_A25	◀▶	GPIO signal
14	33	GPIO_A23	◀▶	GPIO signal
15	34	GPIO_A22	◀▶	GPIO signal
16	35	GPIO_A21	◀▶	GPIO signal
17	36	GPIO_A20	◀▶	GPIO signal
18	37	GPIO_A19	◀▶	GPIO signal
19	38	GPIO_K13	◀▶	GPIO signal
20	39	GPIO_K12	◀▶	GPIO signal
21	40	GPIO_K11	◀▶	GPIO signal
22	41	GPIO_A18	◀▶	GPIO signal
23	42	GPIO_A17	◀▶	GPIO signal
24	43	GPIO_A16	◀▶	GPIO signal
25	44	GPIO_A11	◀▶	GPIO signal
26	45	GPIO_A10	◀▶	GPIO signal
27	46	GPIO_A09	◀▶	GPIO signal
28	47	GPIO_A08	◀▶	GPIO signal

29	48	GPIO_A05	◄►	GPIO signal
30	49	GPIO_A04	◄►	GPIO signal
31	50	GPIO_A03	◄►	GPIO signal
32	51	GPIO_A02	◄►	GPIO signal
33	52	GPIO_A01	◄►	GPIO signal
34	53	GPIO_A00	◄►	GPIO signal
35	GND	DGND	-	Ground
36	GND	DGND	-	Ground

Table 4.13 shows the pin description of J5D100.

Table 4.13 J5D100 Pin Description

Pin Number	J5D100			
	Port Name	Signal Name	DIR	Description
			MCU ◄► J5D100	
1	3.3V	VCP_3P3	-	Power 3.3V
2	3.3V	VCP_3P3	-	Power 3.3V
3	TX0	GPIO_K08	◄►	GPIO signal
4	RX0	GPIO_K01	◄	GPIO signal
5	TX1	GPIO_K09	◄►	GPIO signal
6	RX1	GPIO_K02	◄	GPIO signal
7	TX2	GPIO_K10	◄►	GPIO signal
8	RX2	GPIO_K03	◄	GPIO signal
9	GND	DGND	-	DGND
10	GND	DGND	-	DGND

Figure 4.9 shows the total pin assignment of ten pin headers on the VCP-G.

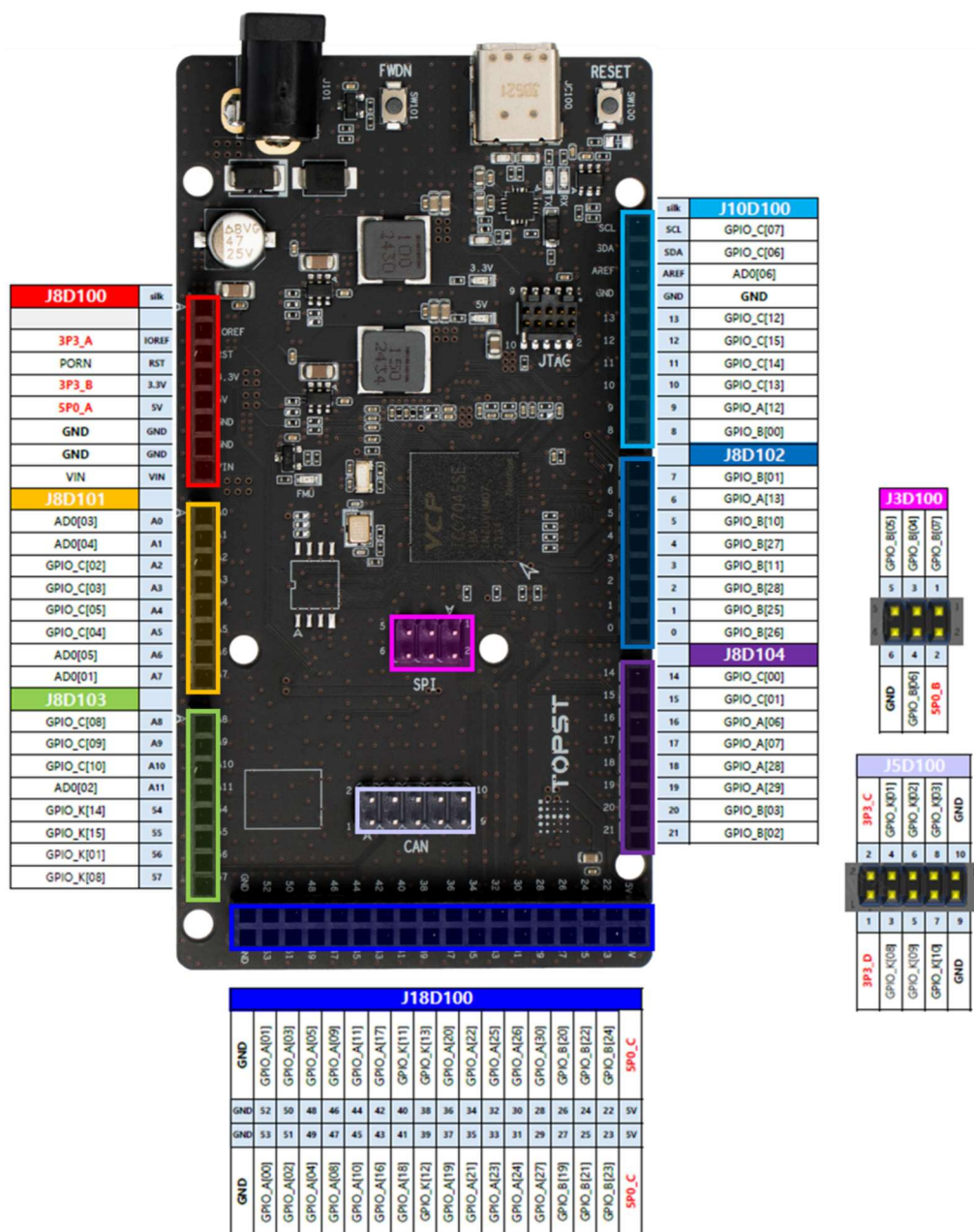


Figure 4.9 Total Pin Assignment of Pin Headers on VCP-G

References

- Contact TOPST for more details: topst@topst.ai

Note: Reference documents can be provided whenever available, depending on the terms of a contract. If the reference documents are unavailable, the contents directly related to your development can be guided.